



9/1/2026

N + - Network Fundamentals & Building Networks.

IPv4 and IPv6.

TASK 1.

- ❖ List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

➤ IPv4 vs IPv6 Addressing:

IPv4	IPv6
Uses 32-bit addresses.	Uses 128-bit addresses.
Written in decimal format , e.g., 192.168.1.10.	Written in hexadecimal format , e.g., 2001:db8::10.
Provides about 4.3 billion unique addresses.	Provides a much larger address space with about 3.4×10^{38} addresses.
Uses ARP to find the MAC address of a device.	Uses Neighbor Discovery Protocol (NDP) instead of ARP.
Supports broadcast communication.	Does not use broadcast ; it uses multicast and anycast.
NAT is commonly used because IPv4 addresses are limited.	NAT is generally not required because of the large address space.

TASK 2.

❖ **Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.**

➤ **Static IPv4 Address Assignment:**

Using the **192.168.10.0/24** network, I assigned a unique static IPv4 address to each home device.

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

TASK 3.

❖ **Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.**

➤ **/27 Subnetting:**

Given: 192.168.1.0/27

✂ Calculation

- ✈ Subnet mask: **255.255.255.224**
- ✈ Borrowed bits: **3**
- ✈ Number of subnets: **$2^3 = 8$**
- ✈ Host bits: **5**
- ✈ Total addresses per subnet: **$2^5 = 32$**
- ✈ Valid hosts per subnet: **$32 - 2 = 30$**
- ✈ Subnet block size: **32**

First Three Subnets

Subnet	Network Address	Broadcast Address	Usable Host IPs
1	192.168.1.0	192.168.1.31	192.168.1.1 – 192.168.1.30
2	192.168.1.32	192.168.1.63	192.168.1.33 – 192.168.1.62
3	192.168.1.64	192.168.1.95	192.168.1.65 – 192.168.1.94

TASK 4.

- ❖ You are given the IP address **10.0.5.17** with subnet mask **255.255.255.0**. Determine the network address, broadcast address, and the range of valid host addresses for this subnet, Convert the subnet mask to CIDR notation and use bitwise **AND** for network address calculation.

- IPv4 Network Calculation:

Given:

✈ IP Address: 10.0.5.17

✈ Subnet Mask: 255.255.255.0

✂ CIDR Notation

255.255.255.0 = /24

✂ Bitwise AND Calculation

IP Address: 10.0.5.17

00001010.00000000.00000101.00010001

Subnet Mask: 255.255.255.0

11111111.11111111.11111111.00000000

AND Result: 10.0.5.0

✂ Results

Item	Answer
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
Valid Host Range	10.0.5.1 – 10.0.5.254
CIDR Notation	/24

TASK 5.

❖ Use an online subnet calculator tool (such as <https://www.subnet-calculator.com/>) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

➤ Subnet Calculator Verification:

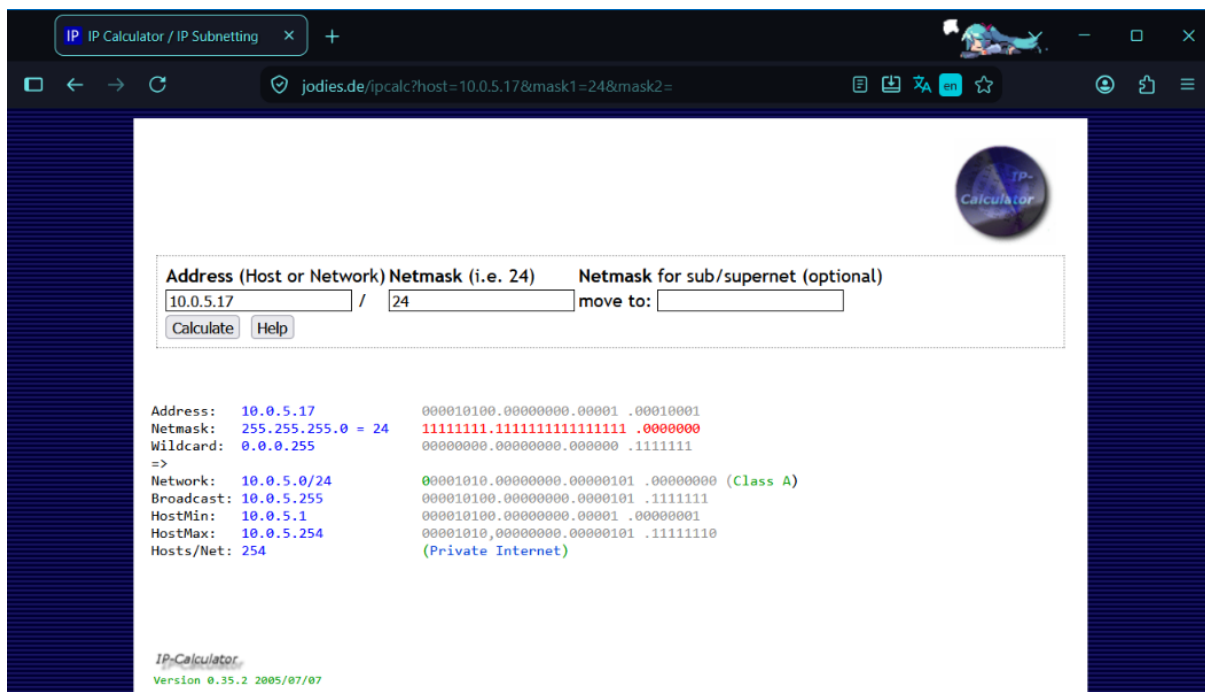


Figure : Subnet Calculator Output for 10.0.5.17/24.

⌘ The calculator shows:

- **Network Address:** 10.0.5.0
- **Broadcast Address:** 10.0.5.255
- **Valid Host Range:** 10.0.5.1 – 10.0.5.254

Verification: The manual calculation matched the subnet calculator's output.